

QTAP: Real-Time Valuation of Ultra-Short Derivative Grids Under Regime-Changing Volatility

Matthew Bui, Cherry Ng, Nhat Tr
AnyAxis Labs
{matthewbui, cherryng, nhattr}@anyax.is

Abstract—While centralized exchanges have matured, standard option workflows predominantly serve participants with long-term time horizons, asking whether a price ends above or below a strike at maturity. The market currently lacks high-frequency derivative mechanisms that offer instantaneous feedback, creating a demand for ultra-short first-passage instruments like tap-trading. Existing models fail to address this because estimating touch probabilities for near-market bands under 1-second observation and over a 5-second loop directly creates mispricing and solvency risk if relying purely on discrete endpoint checks. In this paper, we present QTAP, a quantitative pricing framework designed explicitly for real-time prediction contract infrastructure. To value payouts under these constrained timeframes, we employ a novel method combining Brownian Bridge barrier probability estimation with log-normal Monte Carlo path simulation. To ensure liquidity providers and traders maintain equitable edges during regime-changing volatility, our system incorporates adaptive volatility estimation via EWMA-Parkinson blending and Hawkes-driven jump detection. We evaluate the proposed architecture on chronological crash, quiet, and normal BTC/USDT regimes with 1.8 million discrete OHLCV observations spanning. Empirical validation confirms that our adaptive parameters successfully sustain continuous liquidity and ensure protocol solvency in all tested cases.

Index Terms—High-frequency derivatives, Barrier options, Brownian bridge, Monte Carlo simulation, Adaptive volatility, Hawkes processes, Risk management

I. INTRODUCTION

In recent years, decentralized finance and *high-frequency derivatives* have attracted the attention of both academics and active market participants due to the continuous evolution of digital asset ecosystems. Derivative valuation mechanisms can generally be classified into two main approaches: *long-horizon endpoint contracts* and *ultra-short first-passage instruments* [1], [2]. Although the former relies on standard option workflows that settle at a single maturity date, the latter remains mathematically challenging due to the inherent difficulties of continuous price movement and strict time constraints [3], [4]. Recent methods in market microstructure are moving toward *tap-trading*, where users place orders on whether an asset's price touches a specific band within a compressed sub-minute window [5]. However, pricing these high-frequency contracts introduces a critical bottleneck. Standard discrete simulation at *1-second oracle intervals* fails to capture intra-bar price action, exposing the protocol to unpriced touches. Conversely, models assuming pure continuous monitoring systematically overestimate touch probabilities when applied to discretely observed processes [6]. Under a *1-second observation cadence*, this

trade-off between estimation accuracy and complexity cannot simply average out; it directly translates into *mispricing and solvency risk*.

To remedy these limitations, in this paper, we propose QTAP, a quantitative real-time pricing framework tailored explicitly for micro-horizon prediction contract infrastructure. To tackle the discrepancy between discrete observations and continuous paths, we exploit a *Brownian Bridge Monte Carlo* barrier modeling approach to calculate crossing probabilities between *1-second OHLC* endpoints [3], [7]. QTAP contains a multi-layer pipeline: *adaptive close-to-close volatility estimation*, *Hawkes jump-intensity tracking*, *path simulation*, *touch probability calculation*, and *solvency constraint formulation*. First, to capture both directional persistence and high-frequency wicks, we estimate instantaneous variance using an *Exponentially Weighted Moving Average (EWMA)* blended with a *Parkinson* range-based estimator [8], [9]. Furthermore, to account for the temporal clustering of extreme market moves, we integrate a *Hawkes* self-exciting point process to dynamically update jump intensities [10], [11]. Finally, the features of the simulated paths are aggregated using the *Brownian Bridge barrier lemma* to output an estimated touch probability, and deterministic pseudo-random keys make the *Monte Carlo* output reproducible for the same close price, oracle time, seed, and configuration. In this sense, QTAP is not merely a re-labeling of a classical barrier model; its contribution lies in organizing these ingredients into a continuously quoted tap-trading pipeline under severe time compression. To balance the edge across vastly different asset price levels and stabilize the risk multipliers, we further apply an *adaptive price-level scaling* mechanism. Please note that any price-level scaling is treated as an optional calibration layer rather than as the core pricing theory.

To evaluate our proposed model's performance, we backtest QTAP against real-world, high-frequency historical datasets. We utilize 1.8 million one-second BTC/USDT OHLCV data points, deliberately selected to span three distinct market environments: *crash*, *quiet*, and *normal regimes*. Rather than collapsing these environments into a single aggregate simulation summary, we evaluate the framework through *sampled walk-forward validation*, meaning that calibration uses earlier chronological windows while testing uses later purge-separated windows. The evaluation also includes deterministic execution policies and *feasible-first validation* logic. Then we compare the framework's solvency and multiplier stability

across 30 rigorous scenarios, representing combinations of these 3 market regimes and 10 distinct user order-selection policies. The numerical results demonstrate that our system maintains a balanced operational edge; the framework remains solvent in all evaluated cases.

The remainder of the paper is organized as follows. Section II introduces related works and formulates the prediction grid problem. We describe the details of the proposed *QTAP* methodology, including *volatility estimation* and the *Brownian Bridge pricing pipeline*, in Section III. Section IV presents the experimental design and evaluation results. Finally, we conclude the paper and discuss future implications in Section V.

II. RELATED WORK

The pricing of *barrier options* and derivatives has a rich history in quantitative finance. Comprehensive foundations for option pricing theory and stochastic simulation may be found in the works of Merton [1], Haug [4], and Glasserman [7]. As categorized by existing prediction market designs and option workflows, derivative instruments may be divided into two categories: *long-horizon endpoint contracts* and *continuous first-passage instruments*. Long-horizon contracts typically settle at a single endpoint or maturity date, while *first-passage instruments* depend on whether a price path touches a specific barrier at any point within a defined active time window.

A. Long-horizon and endpoint contracts

Early work mainly focused on endpoint workflows, which ask whether a price ends above or below a strike at maturity, or whether an event occurs by a deadline. In the prediction market space, traditional approaches operate via order books where users trade binary outcome shares. These models often rely on mechanisms like Hanson’s Logarithmic Market Scoring Rule (LMSR) [12], which provided a market maker for thin prediction markets. However, these methods have an inherent limitation, which is operating on days-to-weeks settlement horizons and ignoring intra-bar continuous volatility. While these techniques allow leveraging Central Limit Order Books (CLOB) on-chain, they are insufficient for sub-minute execution loops. Therefore, recent work has shifted focus to ultra-short first-passage instruments, a more challenging problem in stochastic pricing.

B. Ultra-short first-passage instruments

First-passage prediction represents a recent challenging problem in decentralized market infrastructure, which refers to estimating the probability that a price touches a fixed band at any point within a narrow sub-minute window (e.g., 5-30 seconds) using discretely observed *1-second OHLC* data.

The early *continuous barrier models*, formalized by Shreve [3], provide the analytical foundation for computing the probability that a continuous stochastic process crosses a barrier between two observed endpoints. However, continuous barrier models systematically overestimate touch probabilities

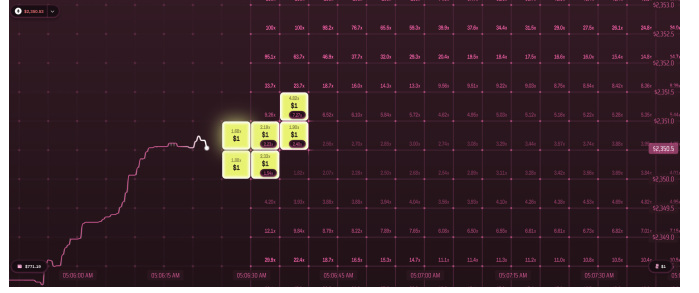


Fig. 1. Tap-trading interface from EuphoriaFi on MegaETH testnet. The displayed market is ETH, not BTC: the visible price ladder spans roughly \$2300–\$2350 with $dP = \$0.5$ bands, and each cell represents a 5-second time window with an associated multiplier.

when applied to discretely observed processes. For time-varying volatility, RiskMetrics [8] popularized the *Exponentially Weighted Moving Average (EWMA)* model; for clustered extreme moves, Bacry et al. [11] demonstrated the effectiveness of *Hawkes self-exciting point processes* [10]. *QTAP* addresses the first-passage problem by combining these adaptive state features with the continuity correction concepts introduced by Broadie and Glasserman [6]. Rather than viewing discrete overestimation as a flaw, our framework leverages this property as a deliberate design choice to ensure protocol solvency under discrete *1-second settlement*.

At the same time, not every classical microstructure direction is well aligned with the object priced by *QTAP*. For example, Roll’s implicit spread estimator [13] is elegant for recovering effective bid–ask spreads from serial covariance in transaction prices, but it is not adopted here because *QTAP* does not seek to infer a latent spread from transaction-by-transaction reversals. Instead, the protocol must estimate *interval-entry probabilities* from 1-second OHLC bars and then map those probabilities into a *solvency-aware quote surface*. That distinction is material: the target object is *first-passage probability*, not spread decomposition.

In the current Web3 product landscape, EuphoriaFi is a notable example of a tap-trading product, currently live on MegaETH testnet. Figure 1 shows the product interface, where users bet on whether the ETH price will touch a specific band within a compressed 5-second cell window. Its public documentation describes Tap Trading as a mobile-first derivatives experience and documents a funded testnet competition, the *Tapathon*, launched on February 16, 2026 [?], [?]. However, the currently available public materials emphasize product positioning, interface flow, and testnet participation details rather than a public mathematical specification of the pricing model or protocol mechanism. Based on those public materials, we therefore do not evaluate EuphoriaFi as an academic modeling contribution at this stage; instead, we treat it as evidence that there is already practical Web3 demand for ultra-short first-passage style products.

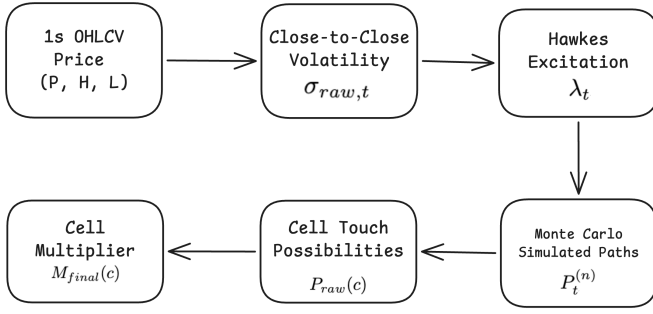


Fig. 2. **QTAP** model flow. The oracle state updates at 1-second cadence; Hawkes intensity enters Monte Carlo through a jump indicator and sampled jump magnitude before probabilities are mapped into multipliers.

III. PROPOSED METHODOLOGY

The **QTAP** framework is comprised of four main interconnected components: **1-second state estimation**, **Monte Carlo path simulation**, **Brownian Bridge touch-probability aggregation**, and **solvency-aware dynamic pricing**. The product-facing contract grid uses fixed price-time cells; in the current principal mode, the price band width is $dP = \$5$, while the oracle state and Monte Carlo path step remain at a **1-second cadence**. The volatility module estimates a close-to-close 1-second state, the **Hawkes** module tracks **jump excitation**, the simulation module converts this state into future price paths, and the pricing module maps raw touch probabilities into bounded multipliers. In the following, we detail these components in Sections III-A, III-B, III-C, and III-D, respectively.

A. Adaptive volatility and jump estimation

In this module, we estimate the **instantaneous volatility state** σ_t from the close price sequence and augment it with a **Hawkes-style excitation state** λ_t for clustered large moves. Given consecutive 1-second close prices P_{t-1} and P_t , the **log return** is

$$r_t = \log \left(\frac{P_t}{P_{t-1}} \right). \quad (1)$$

The classical Parkinson range estimator is [9]

$$v_t^{park} = \frac{[\log(H_t/L_t)]^2}{4 \log 2}. \quad (2)$$

QTAP keeps this formula as a reference and for explicit ablation paths, but disables it in the main **1-second BTC volatility path**. The **observed variance** form is therefore

$$v_t^{obs} = (1 - w_P) r_t^2 + w_P v_t^{park}, \quad w_P = 0, \quad (3)$$

so that

$$v_t^{obs} = r_t^2. \quad (4)$$

This design keeps high/low data in the settlement layer, where it is directly meaningful for proving whether a contract touched a band, while avoiding a range-estimator signal whose 1-second BTC high/low span is often dominated by microstructure noise rather than a robust volatility timescale.

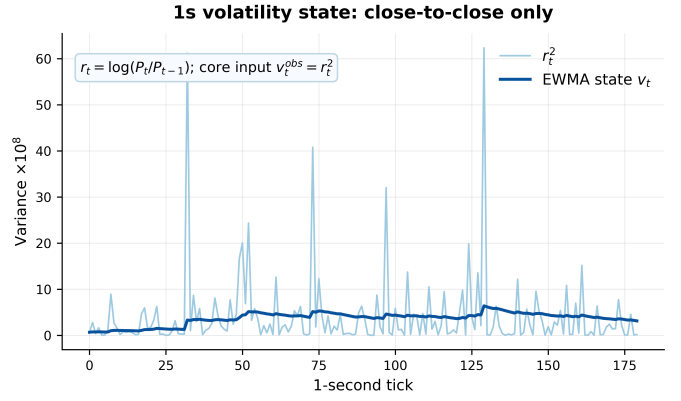


Fig. 3. Observed 1-second close-to-close returns are converted into variance and smoothed by EWMA to produce the volatility state σ_t , which is then reused for both rare-move detection and the diffusion scale of the simulated paths.

We then apply **EWMA memory**,

$$v_t = (1 - \alpha_{vol}) v_t^{obs} + \alpha_{vol} v_{t-1}, \quad (5)$$

where $\alpha_{vol} = \exp(-\log 2/h)$ and h is the half-life in seconds. The volatility state passed forward is

$$\sigma_{raw,t} = \sqrt{v_t}. \quad (6)$$

By default, no sigma-scale or price-level scale is applied in the core path, so $\sigma_{final,t} = \sigma_{raw,t}$. Any scaling path is treated as calibration or ablation, not as the mathematical core. The result of this block is therefore a single **close-based volatility state** that is reused in two places: it normalizes the current tick for rare-move detection, and it sets the diff scale of the forward Monte Carlo paths.

Large cryptocurrency movements exhibit temporal clustering, so **QTAP** also tracks **Hawkes-style excitation**. The return is standardized by the raw volatility state,

$$z_t = \frac{|r_t|}{\sigma_{raw,t}}, \quad (7)$$

then filtered through an adaptive rare-move threshold κ_t :

$$I_t^{obs} = \mathbb{I}\{z_t > \kappa_t\}. \quad (8)$$

Here I_t^{obs} is an observed-tick shock flag, not a simulated future jump. It answers only whether the just-observed 1-second return was rare relative to the current volatility state. That flag then updates the **Hawkes excitation recursion**:

$$\lambda_{t+1} = \mu_0 + \exp(-\beta \cdot \Delta t) \cdot (\lambda_t - \mu_0) + \alpha_{hawkes} \cdot I_t^{obs}. \quad (9)$$

Here μ_0 is the background intensity, α_{hawkes} controls how much an observed rare move excites future jump likelihood, and β controls relaxation back toward baseline. The time index matters: after observing tick t , the engine carries the updated state λ_{t+1} into the forward simulation. At the end of this layer, the state passed forward is $(P_t, \sigma_{final,t}, \lambda_{t+1})$.

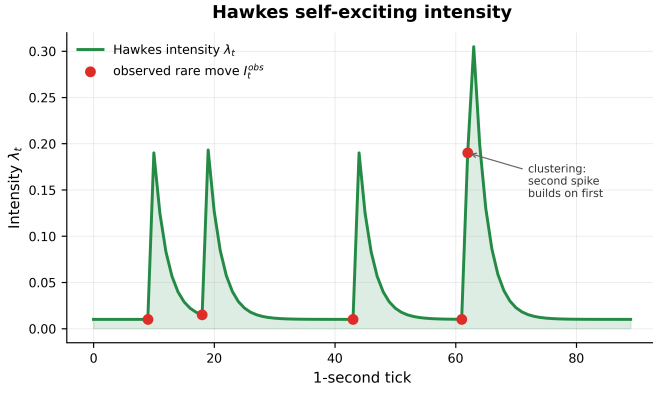


Fig. 4. Hawkes self-exciting intensity λ_t . Each observed rare move (I_t^{obs} , red dots) triggers a spike that decays exponentially toward baseline μ_0 . When two rare moves cluster (ticks 61–62), the second spike builds on the undecayed first, amplifying the jump probability fed into Monte Carlo simulation.

B. Monte Carlo path simulation

We leverage **Monte Carlo** methods to simulate N_{MC} **price paths** over the forward contract horizon [7]. The simulation is stochastic but reproducible: the runtime builds a **deterministic pseudo-random generator** from the configured seed, mode identifier, oracle time index `oracle_second`, and current close price P_t . Player strategies with random probes use a separate deterministic key built from the seed, strategy name, decision tick, and `P_oracle`. Consequently, the same input path, seed, time, close price, and configuration reproduce the same simulated paths and strategy choices.

The path simulator separates three roles. The volatility state $\sigma_{final,t}$ controls the smooth log-return component, the Hawkes state λ_{t+1} controls the probability of a future **Jump** event, and the jump sampler supplies the jump magnitude only when that event is drawn. Hawkes therefore contributes **jump intensity**, not a direct price path. The implementation maps the current intensity to a **one-step jump probability**:

$$q_t = 1 - \exp(-\lambda_{t+1} \cdot \Delta t), \quad (10)$$

and then draws the implemented **MC jump indicator**

$$I_{t,k}^{MC} \sim \text{Bernoulli}(q_t). \quad (11)$$

This is the discrete implementation of the Poisson-process intuition: the probability of at least one jump in the 1-second step is converted into a **Bernoulli event indicator**. The notation intentionally separates I_t^{obs} from $I_{t,k}^{MC}$: the former updates the current Hawkes state, while the latter decides whether simulated path n receives a future jump at step k . If $I_{t,k}^{MC} = 1$, the path samples a jump magnitude $J_{t,k}^{(n)}$ from the bootstrap jump sampler; otherwise the jump contribution is zero.

The smooth part of the path is generated in **log space** so that per-step returns can be added before being exponentiated back into prices. For path n and future step k , draw the **Gaussian shock**

$$\xi_{t,k}^{(n)} \sim \mathcal{N}(0, 1). \quad (12)$$

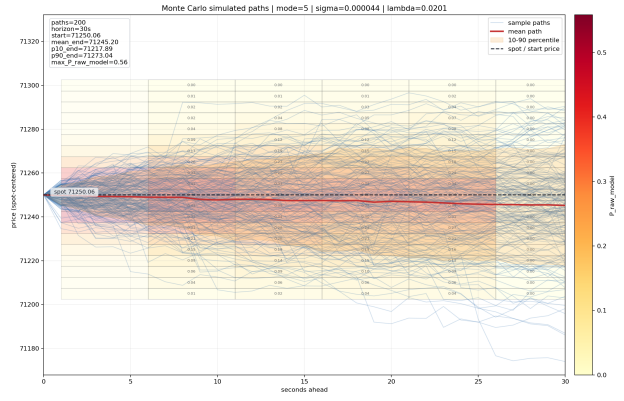


Fig. 5. Representative exact Monte Carlo quote snapshot from the `recent_mar_2026` replay after warmup. The 1-second path fan, 10–90 percentile band, and overlaid cell-level P_{raw} values illustrate how path-to-cell aggregation produces a mode-5 probability surface before multiplier construction.

The **diff return component** is

$$r_{t,k}^{diff,(n)} = -\frac{1}{2} \cdot \sigma_{final,t}^2 \cdot \Delta t + \sigma_{final,t} \cdot \sqrt{\Delta t} \cdot \xi_{t,k}^{(n)}. \quad (13)$$

The first term is the **Itô drag** required for zero-drift log-normal diffusion [3], [14]; it prevents the exponentiated path from developing an artificial upward bias. The full log-space increment is

$$\Delta \ell_{t,k}^{(n)} = r_{t,k}^{diff,(n)} + I_{t,k}^{MC} \cdot J_{t,k}^{(n)}. \quad (14)$$

Thus $\Delta \ell_{t,k}^{(n)}$ is the one-step simulated log return: it is the sum of the smooth diff term and the conditional jump term. The **simulated price path** is then recovered by cumulating these log increments:

$$P_{t+k}^{(n)} = P_t \cdot \exp \left(\sum_{j=1}^k \Delta \ell_{t,j}^{(n)} \right). \quad (15)$$

The current implementation keeps $\sigma_{final,t}$ and λ_{t+1} fixed across the short MC horizon. This is a deliberate short-horizon approximation for speed and stability, not a claim that the future Hawkes state is evolved step-by-step inside the simulated path. Figure 5 shows a representative **exact Monte Carlo quote snapshot** from the `recent_mar_2026` regime after warmup. Instead of a schematic toy example, the figure uses the actual March 2026 replay state so the 1-second path fan, percentile band, and overlaid cell-level probabilities can be read directly against the mode-5 quote surface.

C. Brownian Bridge barrier probability

The core mathematical difficulty lies in the gap between **continuous-time price motion** and **discrete-time observation**. Settlement uses historical 1-second high/low data over the locked contract window, while the pricing engine must estimate whether a future path segment may enter a price band between adjacent simulated endpoints. We solve this using the **Brownian Bridge Barrier Lemma** [3].

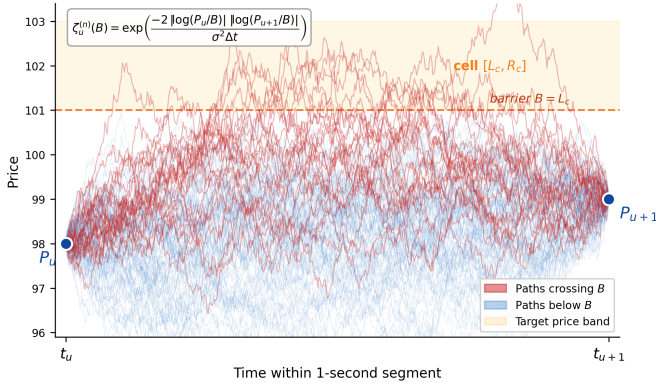


Fig. 6. Brownian Bridge barrier-crossing probability. Both endpoints P_u, P_{u+1} lie below barrier $B = L_c$; red paths cross the barrier while blue paths remain below. A discrete endpoint check would report zero touch probability, but $\zeta > 0$ captures the continuous crossing risk.

For a cell c , let $[L_c, R_c]$ be the quoted price band and let $\mathcal{W}(c)$ be the cell's 5-second time window. The index $u \in \mathcal{W}(c)$ denotes one 1-second simulated segment inside that cell window, with endpoints $P_u^{(n)}$ and $P_{u+1}^{(n)}$ for path n . Given a barrier level B , the **Brownian Bridge touch probability** for that single segment is

$$\zeta_u^{(n)}(B) = \exp\left(-\frac{2 \cdot \left| \log(P_u^{(n)}/B) \right| \cdot \left| \log(P_{u+1}^{(n)}/B) \right|}{\sigma_{final,t}^2 \cdot \Delta t}\right), \quad (16)$$

when both endpoints lie on the same side of B . This is the same distance-penalty form as the pipeline expression, written directly in terms of the two log-distances to avoid introducing helper symbols in the paper. Figure 6 illustrates the mechanism: even when both endpoints lie below the barrier, a fraction of continuous bridge paths cross it. Without the Brownian Bridge correction, a discrete endpoint check would report zero touch probability for all such segments.

For the interval cell c , let $\mathcal{B}_c = [L_c, R_c]$. The relevant barrier depends on where the two segment endpoints sit relative to $\mathcal{B}_c$:

$$P_{touch,u}^{(n)}(c) = \begin{cases} 1, & P_u^{(n)} \in \mathcal{B}_c \text{ or } P_{u+1}^{(n)} \in \mathcal{B}_c, \\ \zeta_u^{(n)}(L_c), & P_u^{(n)}, P_{u+1}^{(n)} < L_c, \\ \zeta_u^{(n)}(R_c), & P_u^{(n)}, P_{u+1}^{(n)} > R_c. \end{cases} \quad (17)$$

Thus $P_{touch,u}^{(n)}(c)$ is the probability that path n touches cell c during the 1-second segment u . The full 5-second cell probability is one minus the probability of missing the cell in every segment of $\mathcal{W}(c)$:

$$p^{(n)}(c) = 1 - \prod_{u \in \mathcal{W}(c)} (1 - P_{touch,u}^{(n)}(c)). \quad (18)$$

Finally, the **raw contract probability** is the Monte Carlo average across paths,

$$P_{raw}(c) = \frac{1}{N_{MC}} \sum_{n=1}^{N_{MC}} p^{(n)}(c). \quad (19)$$

D. Pricing formulation and risk controls

To guarantee protocol solvency, the raw estimated probability $P_{raw}(c)$ undergoes a multi-layer pricing pipeline. In the paper notation, uppercase M is reserved for user-facing multipliers, while lowercase m denotes internal margin fractions. The **fair multiplier** before protocol margin is

$$M_{fair}(c) = \frac{1}{P_{raw}(c)}. \quad (20)$$

The **effective margin fraction** combines a baseline margin and dynamic risk premium:

$$m_{eff} = m_{base} + \min\left(a_\sigma \max\left(0, \frac{\sigma_{final,t}}{\sigma_{ref}} - 1\right) + a_\lambda \max\left(0, \frac{\lambda_{t+1}}{\lambda_{ref}} - 1\right), m_{risk,max}\right). \quad (21)$$

This keeps the volatility and Hawkes terms dimensionless by normalizing them against reference levels before applying the sensitivity weights a_σ and a_λ . The reference levels are **calibration baselines** for BTC/USDT 1-second operation, not structural constants. As a scale check, a BTC annualized volatility near 50% corresponds under square-root-of-time scaling to a total 1-second volatility of approximately $0.50/\sqrt{31,536,000} \approx 8.9 \cdot 10^{-5}$. **QTAP** then uses a close-to-close **EWMA volatility state** while representing rare-move clustering through the **Hawkes layer**, so the operational reference is set lower: the local calibration sweep selects $\sigma_{ref} = 3 \cdot 10^{-5}$ to keep $\sigma_{final,t}/\sigma_{ref}$ active without saturating the dynamic margin term. On the jump side, $\lambda_{ref} = 0.1 \text{ s}^{-1}$ is the reference excitation level; under $q = 1 - \exp(-\lambda \Delta t)$, it corresponds to one expected jump per ten 1-second bars and $q \approx 9.5\%$ for $\Delta t = 1$ second.

The **margin-adjusted multiplier** is therefore

$$M_{eff}(c) = (1 - m_{eff}) \cdot M_{fair}(c) = \frac{1 - m_{eff}}{P_{raw}(c)}. \quad (22)$$

This is the last probability-only quote before inventory and solvency overlays.

Risk management then maps M_{eff} into the final quoted multiplier. First, an **inventory penalty** reduces the quote as cell-level liability $L_{used}(c)$ grows relative to protocol capacity C_{pool} :

$$K_{skew}(c) = \frac{1}{1 + \beta_{skew} \cdot L_{used}(c)/C_{pool}}. \quad (23)$$

Second, a **tail-risk safety factor** compares the simulated payout-loss distribution $\mathcal{L}$ against the protocol risk budget:

$$K_{safety} = \min\left(1, \frac{R_{budget}}{CVaR_\alpha(\mathcal{L})}\right). \quad (24)$$

The **final multiplier** is

$$M_{final}(c) = \text{clamp}\left(M_{eff}(c) \cdot K_{skew}(c) \cdot K_{safety}, M_{min}, M_{max}\right). \quad (25)$$

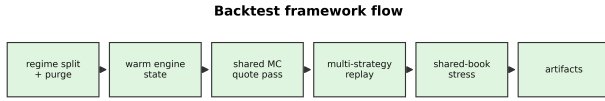


Fig. 7. Backtest framework flow. The framework runs a shared Monte Carlo quote pass for each regime, evaluates multiple strategy states on that same market path, and reserves shared-book stress for finalist or higher-Monte-Carlo artifacts.

The numerical choices for β_{skew} , C_{pool} , R_{budget} , α , M_{min} , and M_{max} are not universal model constants and are not recommendations made by this paper. They are **protocol policy inputs** determined by available capital, liquidity depth, operating constraints, and risk tolerance. Price-level scaling, if enabled in an experiment, is treated as a calibration layer outside the core theory-first path.

IV. PERFORMANCE EVALUATION

This section evaluates **QTAP** as a **sequential pricing framework** rather than as a stand-alone classifier. The audit follows the unified **backtest pipeline** used by the current framework:

- 1s OHLCV $\rightarrow$ chronological split $\rightarrow$ purge/warmup
 $\rightarrow$ shared MC quote pass $\rightarrow$ strategy replay
 $\rightarrow$ forecast, surface, and feasibility gates. (26)

We place the main evidence first: the **backtest artifact**, **calibration diagnostics**, and **quote-surface health**. Shared-book stress and liability screens remain part of the feasible-first engineering gate, but the paper-facing analysis focuses on **probability calibration**, **realized touch behavior**, and **multiplier distribution**.

A. Evaluation protocol and datasets

The experiment uses BTC/USDT **1-second OHLCV regimes** selected to cover crash, quiet, and recent market conditions. Each **regime** is split in chronological order, with a **60/20/20 screening split** and a **70-second purge/embargo gap** around validation and test boundaries. The purge exceeds the **65-second maximum label horizon** used by the current grid settlement, reducing leakage from overlapping future labels [15].

The evaluation artifact replays **2000 MC paths**, **10000 probe ticks**, and **10 strategy probes** per regime. The three held-out probe windows are:

- `crash_oct_2025`: 2025-10-13 14:26:19–17:12:58 UTC
- `quiet_jan_2026`: 2026-01-06 14:26:19–17:12:58 UTC
- `recent_mar_2026`: 2026-03-31 14:26:19–17:12:58 UTC

Warmup is not treated as training. The engine replays pre-test data before the probe window so that **close-to-close**

TABLE I
STRATEGY PROBES REPLAYED IN THE EVALUATION ARTIFACT.

Strategy	Class	Selection rule
<code>closest_band</code>	Nearest-cell	Nearest price band at nearest valid horizon.
<code>col_sweep_nearest</code>	Column sweep	All price bands in the nearest valid time column.
<code>row_sweep</code>	Row sweep	All time windows for the price row closest to spot.
<code>three_rows</code>	Local sweep	Spot row and adjacent rows at nearest valid horizon.
<code>long_bias</code>	Directional	Price bands above spot, ordered by distance and time.
<code>short_bias</code>	Directional	Price bands below spot, ordered by distance and time.
<code>vol_spike_nearest</code>	Vol. heuristic	Nearest-column sweep when the volatility threshold is active.
<code>momentum_nearest</code>	Jump heuristic	Nearest-column cell shifted upward when the jump flag is active.
<code>mean_revert_nearest</code>	Jump heuristic	Nearest-column cell shifted downward when the jump flag is active.
<code>new_grid_full</code>	Diag. sweep	Full-grid probe when a new grid opens.

volatility memory, **Hawkes excitation**, **calibration factors**, and **oracle state** are initialized from the past rather than from an artificial first evaluation tick. The evaluated product surface is **Mode 5**, where the public price band width is $dP = \$5$. Oracle updates and Monte Carlo path steps remain at the **1-second cadence**, and settlement is checked against historical **1-second high/low evidence** over the locked contract window.

B. Framework execution

The current evaluation path is the unified **multi-strategy framework**. For each regime, the framework updates the oracle at 1-second cadence and produces a **direct per-tick quote stream** from the engine. Canonical strategy replay then consumes the **exact quote tape** generated for that candidate-regime unit. This avoids repeated quote generation while keeping every reported strategy on the same market path, warmup state, quote stream, and settlement rule.

The framework now includes a disk-backed **exact quote tape** for fixed-quote replay. Parameter values, probability surfaces, multiplier surfaces, and quote tapes remain specific to the evaluated configuration. Liability-sensitive shared-book stress remains on a separate exact path and is not represented as a replay of the fixed quote tape.

C. Strategy probes

The artifact replays all ten **strategy probes** listed in Table I. They are not claims about a representative user population. They are controlled probes that consume the same quote stream without look-ahead and expose how different cell-selection rules sample the **probability surface** and **multiplier surface**.

D. Metrics and gates

The evaluation separates **forecast quality** from **protocol feasibility**. Forecast quality is measured using **Brier score** and **Expected Calibration Error (ECE)**:

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (P_{raw}(c_i) - y_i)^2, \quad (27)$$

where $y_i \in \{0, 1\}$ is the realized settlement outcome. **Brier** is a strictly proper scoring rule for probabilistic forecasts [16], [17]. It combines calibration and sharpness into a single measure: 0 is perfect, and 0.25 is the conventional reference point for a naive 50/50 forecast on balanced binary labels.

TABLE II
REGIME-LEVEL CALIBRATION DIAGNOSTICS.

Regime	Mean P_{hit}	Realized hit	Brier	vs. 0.25	ECE
crash_oct_2025	19.74%	16.34%	0.1366	−45.4%	4.21 pp
quiet_jan_2026	19.35%	17.29%	0.1428	−42.9%	3.90 pp
recent_mar_2026	19.01%	16.77%	0.1385	−44.6%	4.28 pp

ECE summarizes the bucket-level gap between empirical hit frequency and predicted confidence [18]:

$$ECE = \sum_{b=1}^B \frac{|I_b|}{N} |\text{acc}(I_b) - \text{conf}(I_b)|. \quad (28)$$

ECE is zero under perfect bucket-level calibration. Because ultra-short one-touch cells are not a standard binary-classification benchmark, we use these score semantics as standard forecasting diagnostics rather than as universal acceptance thresholds. We therefore interpret **single-digit ECE** as the practical target range, while **double-digit ECE** signals a material mismatch between quoted and realized touch frequency.

Feasibility gates complement forecast quality with quote-surface health. M_{final} is hard-clamped to $[1.05 \times, 25 \times]$, but the preferred operating band is $1.2 \times - 8 \times$ because it balances user interest with tail-risk control. Floor-compression diagnostics track whether any cell in a 120-cell snapshot approaches the $1.05 \times$ hard floor and how much of the board is pinned to the minimum payout. Operating-band diagnostics measure the surface area that remains within the preferred $1.2 \times - 8 \times$ range.

E. Results

Table II reports regime-level calibration diagnostics. All three regimes achieve **Brier scores** between 0.13 and 0.15, representing a **42–45% improvement** over the naive 0.25 baseline. ECE remains in the **3.9–4.3 pp** range, well within the single-digit practical target.

Quote-surface feasibility confirms that floor compression remains contained: the fraction of snapshots with any cell at $M_{final} \leq 1.10$ stays below **8%** in the crash and quiet regimes, rising to **17.7%** in the more volatile recent regime. The preferred $1.2 \times - 8 \times$ operating band captures **54–62%** of the 120-cell surface across all three regimes. A representative exact quote snapshot from the `recent_mar_2026` replay is shown in Figure 5.

Table III reports strategy-level calibration and hit behavior for the full tested strategy set; mean P_{hit} and Brier/ECE are weighted by bet count. The results are intentionally centered on P_{hit} , **realized hit frequency**, **Brier**, **ECE**, and **average multiplier** rather than on per-strategy economic return.

Table IV shows how each strategy samples the **multiplier surface**. Buckets are computed on the actual cells selected by each probe, not on the full quote surface. Nearest-cell policies concentrate in the **low-multiplier buckets** because they repeatedly choose higher-probability cells. Broad column, directional, and full-grid sweeps spend more mass in the $2 \times -$, $5 \times$ and $5 \times +$ regions.

TABLE III
STRATEGY-LEVEL CALIBRATION ACROSS THREE PROBE WINDOWS.

Strategy	P_{hit}	Real. hit	Brier	ECE	M_{final}
closest_band	33.99%	33.35%	0.2330	10.03	2.06
col_sweep_nearest	18.82%	16.00%	0.1302	3.27	3.91
row_sweep	28.93%	25.09%	0.1996	9.60	2.59
three_rows	33.75%	30.46%	0.2238	9.98	1.98
long_bias	17.89%	15.87%	0.1344	3.47	4.01
short_bias	17.91%	16.25%	0.1376	3.29	4.00
vol_spike_nearest	18.85%	16.01%	0.1303	3.25	3.89
momentum_nearest	33.99%	33.34%	0.2329	10.02	2.06
mean_revert_nearest	33.99%	33.35%	0.2330	10.02	2.06
new_grid_full	19.37%	14.43%	0.1285	6.04	5.02

TABLE IV
PLACED-BET MULTIPLIER DISTRIBUTION BY STRATEGY. THE FIRST NUMERIC COLUMN REPORTS M_{final} AS MEAN $\pm$ STANDARD DEVIATION, POOLED ACROSS ALL PLACED BETS FROM THE THREE PROBE WINDOWS USING THE PER-REGIME BET COUNT, MEAN, AND STANDARD DEVIATION.

Strategy	M_{final} (mean $\pm$ std)	[1, 1.5]	[1.5, 2]	[2, 5]	5+
closest_band	2.06 $\pm$ 1.65	47.9%	26.4%	21.3%	4.5%
col_sweep_nearest	3.91 $\pm$ 3.37	10.1%	13.6%	58.5%	17.9%
row_sweep	2.59 $\pm$ 1.58	17.8%	23.8%	52.1%	6.3%
three_rows	1.98 $\pm$ 1.46	44.6%	30.5%	21.7%	3.3%
long_bias	4.01 $\pm$ 2.97	6.3%	10.6%	61.9%	21.2%
short_bias	4.00 $\pm$ 2.97	6.3%	10.7%	62.0%	21.1%
vol_spike_nearest	3.89 $\pm$ 3.32	10.1%	13.6%	58.7%	17.6%
momentum_nearest	2.06 $\pm$ 1.67	47.9%	26.4%	21.3%	4.5%
mean_revert_nearest	2.06 $\pm$ 1.67	47.9%	26.4%	21.3%	4.5%
new_grid_full	5.02 $\pm$ 4.02	5.6%	9.0%	52.0%	33.4%

The validated comparison artifact does not include shared-book stress rows. We therefore omit numeric stress claims here and reserve **shared-book liability** tables for finalist or higher-Monte-Carlo artifacts that explicitly run the **stress stream**.

V. CONCLUSION

In this paper, we proposed *QTAP*, a real-time quantitative pricing framework for *ultra-short derivative contracts*, which integrates *close-to-close 1-second volatility estimation*, *Hawkes jump-intensity tracking*, *Brownian Bridge interval-hit probability*, and *solvency-aware multiplier construction*. The framework keeps the *oracle state*, *Monte Carlo path step*, and *settlement evidence* aligned with 1-second OHLCV data while preserving fixed product-facing price–time cells. At the same time, the evaluation protocol makes clear that *forecast quality*, *strategy outcomes*, *quote-surface compression*, and *shared-book stress* must be interpreted as separate evidence streams rather than collapsed into a single average margin. In summary, *QTAP* demonstrates that *ultra-short first-passage valuation* can be organized as a reproducible pipeline from *1-second Close and Time* anchored observations to solvency-aware quotations under severe time compression.

In future work, we plan to validate finalists with *higher-Monte-Carlo artifacts*, stress *shared-book liabilities* more broadly, and evaluate whether richer forward-state dynamics for σ and λ improve short-horizon pricing without sacrificing reproducibility.

ACKNOWLEDGMENT

We would like to thank AnyAxis Labs, Holdstation for supporting our research.

REFERENCES

- [1] R. C. Merton, "Theory of rational option pricing," *Bell Journal of Economics*, vol. 4, no. 1, pp. 141–183, 1973.
- [2] F. Black and M. Scholes, "The pricing of options and corporate liabilities," *Journal of Political Economy*, vol. 81, no. 3, pp. 637–654, 1973.
- [3] S. E. Shreve, *Stochastic Calculus for Finance II: Continuous-Time Models*. Springer, 2004.
- [4] E. G. Haug, *The Complete Guide to Option Pricing Formulas*, 2nd ed. McGraw-Hill, 2007.
- [5] Á. Cartea, S. Jaimungal, and J. Penalva, *Algorithmic and High-Frequency Trading*. Cambridge University Press, 2015.
- [6] M. Broadie and P. Glasserman, "A continuity correction for discrete barrier options," *Mathematical Finance*, vol. 7, no. 4, pp. 325–349, 1997.
- [7] P. Glasserman, *Monte Carlo Methods in Financial Engineering*. Springer, 2004.
- [8] J.P. Morgan/Reuters, "Riskmetrics technical document," J.P. Morgan, Tech. Rep., 1996.
- [9] M. Parkinson, "The extreme value method for estimating the variance of the rate of return," *Journal of Business*, vol. 53, no. 1, pp. 61–65, 1980.
- [10] A. G. Hawkes, "Spectra of some self-exciting and mutually exciting point processes," *Biometrika*, vol. 58, no. 1, pp. 83–90, 1971.
- [11] E. Bacry, I. Mastromatteo, and J.-F. Muzy, "Hawkes processes in finance," *Market Microstructure and Liquidity*, vol. 1, no. 1, 2015.
- [12] R. Hanson, "Combinatorial information market design," *Information Systems Frontiers*, vol. 5, no. 1, pp. 107–119, 2003.
- [13] R. Roll, "A simple implicit measure of the effective bid-ask spread in an efficient market," *The Journal of Finance*, vol. 39, no. 4, pp. 1127–1139, 1984.
- [14] B. Oksendal, *Stochastic Differential Equations: An Introduction with Applications*, 6th ed. Springer, 2003.
- [15] M. López de Prado, *Advances in Financial Machine Learning*. Wiley, 2018.
- [16] G. W. Brier, "Verification of forecasts expressed in terms of probability," *Monthly Weather Review*, vol. 78, no. 1, pp. 1–3, 1950.
- [17] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *Journal of the American Statistical Association*, vol. 102, no. 477, pp. 359–378, 2007.
- [18] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning*, 2017, pp. 1321–1330.